

A modified multiplicative updates algorithm for KL-NMF

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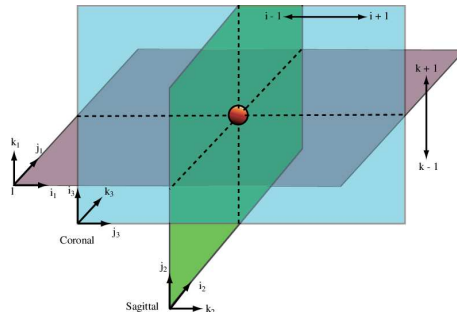
1 Introduction

Recall NMF consists in modelling $V \simeq WH$, with $W \in \mathbb{R}_+^{F \times K}$ and $H \in \mathbb{R}_+^{K \times N}$.

The undercomplete case ($K < \min(F, N)$) has been the object of extensive investigation Lee and Seung (1999); Cichocki et al. (2006a); Laurberg et al. (2008); Lin (2007); Badeau et al. (2011); Smaragdis and Brown (2003); Gillis (2020)

However, while sparse linear modelling has been pushing the barriers of dictionary sizes so far as to allow entire libraries of dictionaries to be fed to linear modelling, little has been published in the NMF community about sparse coding with overcomplete dictionaries. In (Smaragdis et al., 2009), the authors use the entire training data as a dictionary and propose sparse coding for a speech separation task. While sensible from an audio signal processing point of view, this approach induces an extremely expensive coding step. Therefore, the question of learning an overcomplete dictionary in NMF is left open.

We argue in this study that this question can be solved, starting from geometric intuition : the number of atoms K can be interpreted as the number of vertices of a polytope seen in perspective. This is different from linear modelling where K refers the the dimension of linear vector space. While overcomplete sparse modelling leads to deal with a complex nonconvex manifold, overcomplete conic modelling simply leads to study polytopes with an increasing number of vertices.



We will frequently talk about a cone and its shadow. Imagine for instance that we have a zero-pointed cone in 3d space. Consider the perspective plane defined by $\mathcal{P} = \{(x, y, z) \in \mathbb{R}^3, x + y + z = 1\}$.

Then the intersection of $\mathcal{P}$ with our cone is simply a polygon in two dimensions. This metaphor will be useful to introduce various notions about conic modelling that we envisage in this study.

A COUPLE OF NICE FIGURES ARE NEEDED HERE TO EXPLAIN

There are two main problems in devising an overcomplete dictionary learning procedure with NMF : the first problem is that of handling non-euclidean loss function : an early proposal was the multiplicative updates (Lee and Seung, 1999; Cao et al., 1999). The convergence analysis of this algorithm, in the undercomplete case, shows that under certain conditions, linear convergence may be achieved. However, some modifications must be brought to avoid the problem of absorbing zeroes.

Finding efficient algorithms to optimize the KL divergence, and more generally the family of beta-divergences, is still a research track and many propositions have been made (Cichocki et al., 2006b; Sun and Fevotte, 2014; Bauschke et al., 2017; Yanez and Bach, 2017; Hien and Gillis, 2021; Perraudin et al., 2024) However, to the best of our knowledge, either they rely on the hypothesis that all activation coefficients are nonnegative, or they fail to exhibit linear convergence rates. This becomes a real problem in the overcomplete case, where many zeroes are expected not only in the final estimate, but also during the optimization : for instance, one might expect that the activation coefficients associated with two close words of the dictionary will oscillate around zero at some iterations.

The third and last problem is to deal with the potential nonconvexity of the problem when both the dictionary and the activation coefficients are optimized. In the undercomplete case, the volume constraint has been proposed (Ang and Gillis, 2019).

In this study, we propose a modified MU algorithm, based on a modified surrogate function. Furthermore, we propose several penalties for the size of the cone, in order to deal with the nonconvexity of the problem.

2 Notations

2.1 linear algebra and optimization

TODO

2.2 The volume of a F dimensional cone with $K + 1$ vertices

The simplex of dimension K is $\mathbb{S}_K = \{h \in \mathbb{R}^K, h_k \geq 0, \sum_k h_k = 1\}$

Given $W \in \mathbb{R}^{F \times K} = (\vec{w}_1, \dots, \vec{w}_k)$ we will refer to $\mathcal{C}(W) = W\mathcal{S}_K$, so that $\mathcal{C}(W)$ is a closed bounded polytope.

In the case $F = K$, the volume of $\mathcal{C}(W)$ is equal to $\text{vol}(W) = \frac{1}{F!} \det(W)$. If $F < K$, the volume of $\mathcal{C}(W)$ is still defined in the vector space $W\mathbb{R}^K$, and it can be computed as $\text{vol}(W) = \sqrt{\det(W'W)}$.

However, if $K > F$, the volume of $\mathcal{C}(W)$ must be obtained by simplicial decomposition, so there is no closed-form formula.

3 Supervised KL-NMF : model and algorithm

Given an array of data $V \in \mathbb{R}_+^{F \times N}$, nonnegative matrix factorization (NMF) consists in finding a dictionary matrix $W \in \mathbb{R}_+^{F \times K}$ and a decomposition matrix $H \in \mathbb{R}_+^{K \times N}$ such that $V \simeq WH$. If a strict equality constraint is imposed, then we talk about exact NMF. Matrices W and H are required to have nonnegative coefficients, that is $\forall f, k, W_{fk} \geq 0$ and $\forall k, n, H_{kn} \geq 0$.

In many applications, exact factorization is not possible: there might be extra noise during acquisition of the data, of the data itself may be sampled from an underlying probabilistic model. For these reasons, one is led to solve NMF as an optimization problem of the following form :

$$\begin{aligned} \min \quad & D(V, WH) \\ & W \geq 0 \\ & H \geq 0 \end{aligned} \tag{1}$$

where D measures the difference between the observed data V and the estimated model $\hat{V} = WH$.

When the dictionary matrix W is known, but H is unknown, we will talk about supervised NMF, like in (Badeau et al., 2011).

3.1 Divergence measures and the Kullack-Leibler divergence

Given arrays $X, Y \in \mathbb{R}_+^{F \times N}$

$$D(Y, X) = \sum_{f=1}^F \sum_{n=1}^N d(y_{fn}, x_{fn})$$

The divergence measure we will study in this manuscript is the Kullback-Leibler divergence.

$$d_{KL}(y, x) = -y \ln(x/y) + x - y$$

As long as the context makes it clear, we will write $d(y, x)$ instead of $d_{KL}(y, x)$ throughout this manuscript.

The Kullback-Leibler (KL) divergence is defined as an extended-value function on $\mathbb{R}_+^2$, with the convention that $d(0, x) = d(y, 0) = 0$ for positive x and y , and $d_{KL}(0, 0) = 0$.

The KL divergence is convex as a function of x and as a function of y .

More precisely, $d'_x(y, x) = -\frac{y}{x} + 1$, $d'_y(y, x) = \ln y - \ln x$, and

$$\nabla^2 d(y, x) = \begin{pmatrix} 1/y & -1/x \\ -1/x & y/x^2 \end{pmatrix}$$

Notice that the trace of the Hessian is equal to $1/y + y/x^2$ and its determinant to $2/x^2$, so the Hessian is positive definite as long as $x > 0$ and $y > 0$. Therefore, the KL divergence is convex on the interior of its domain : $\text{int}(\text{dom}d_{KL}) = \{(x, y) \in \mathbb{R}^2, x > 0, y > 0\}$. One can check that it is convex on its whole domain.

Other common measures of divergence : the family of β -divergences in particular encompasses the quadratic loss, the KL divergence (used in audio and image processing), and the IS divergence (used in audio processing).

Among those, the KL divergence has the property of being more sensitive to scale than the IS divergence, but less than the quadratic loss. It appears as a loss function in several statistical models : the multinomial model (Smaragdis and Brown, 2003) and the poisson model (Virtanen et al., 2008).

3.2 One block of the algorithm : Poisson linear modelling

The loss function is separable with respect to the columns of matrix H : any algorithm for NMF will thus include a subroutine to solve the following optimization problem :

$$\begin{aligned} \min_{h \geq 0} \quad & D_{KL}(v, Wh) \end{aligned} \tag{2}$$

where $h \in \mathbb{R}_+^K$. The inequality $h \geq 0$ means that $\forall k, h_k \geq 0$. Define the loss function of this problem $l(h)$.

The domain of l is $\text{dom}(l) = \{\}$. It is non-empty under the following condition :

$$\forall f, v_f > 0 \Rightarrow \exists k, W_{fk} > 0$$

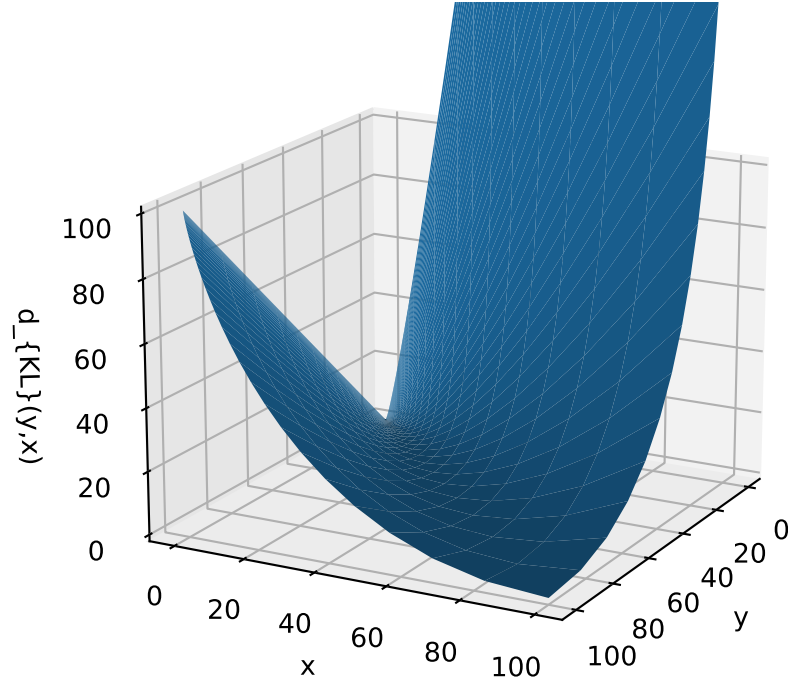


Figure 1: Surface plot of the KL-divergence as a two-variable function

l is convex. It is differentiable on $\text{dom}(f)$ and its first order derivatives are :

$$\partial_k l(h) = \sum_f W_{fk} \left(-\frac{v_f}{\hat{v}_f} + 1 \right)$$

where $\hat{v} = Wh$.

KKT conditions of optimality Let $h^* \geq 0$ be a local minimum of l . Then, for all $k \in \{1, \dots, K\}$, one of the following conditions is true :

- $h_k^* > 0$ and $\partial_k l(h^*) = 0$
- $h_k^* = 0$ and $\partial_k l(h^*) \geq 0$

We propose the term "conic modelling" in contrast with "linear modelling", to stress the fact that, although optimization in vector spaces accomodates linear constraints such as nonnegativity constraints, using a routine operation such as subtraction is to be avoided as much as possible, and possibly finding algebraic operations adapted to this geometry would be preferable to standard vector geometry (Aitchison, 1982).

3.3 Auxiliary functions and MM algorithms

Definition 1 U is an auxiliary function of l at $\underline{h}$ if the following conditions are met :

- $\forall h, l(h) \leq U(h)$
- $l(\underline{h}) = U(\underline{h})$

Property 2 The following are necessary conditions for U to be an auxiliary function of l at $\underline{h}$:

- $l(\underline{h}) = U(\underline{h})$
- $\nabla l(\underline{h}) = \nabla U(\underline{h})$

Property 3 A sufficient condition for U to be an auxiliary function of l at $\underline{h}$ is that $U - l$ is convex and that $\underline{h}$ is a minimum of $U - l$.

4 Existing algorithms

In order to find a solution of the KL conic regression problem, algorithms must be devised since the set of KKT inequations is intractable.

Known algorithms to solve supervised KL-NMF : multiplicative updates (MU), block-mirror descent (BMD) (Hien and Gillis, 2021), CCD (Hsieh and Dhillon, 2011), scaled Newton (SN) (Hien and Gillis, 2021), primal-dual PD (Yanez and Bach, 2017). CCD obtains interesting performances, as well as SN, but the multiplicative updates remains a good candidate in most situations.

We will discuss the approach introduced in (Hien and Gillis, 2021) in more details in Section 4.2

4.1 The multiplicative updates algorithm

(Lee and Seung, 1999) use the Jensen inequality to majorize the log term of the objective function and obtain the well-known multiplicative updates algorithm :

$$l(h) \leq U(h, \underline{h}) = \sum_k -z_k \log(h_k) + q_k h_k$$

where $p_k = \sum_f W_{fk} \frac{v_f}{\hat{v}_f}$ and $q_k = \sum_f W_{fk}$.

Minimizing U yields the multicative updates iteration :

$$h_k \leftarrow h_k \frac{\sum_f W_{fk} \frac{v_f}{\hat{v}_f}}{\sum_f W_{fk}}$$

Theorem 4 *Convergence of the multiplicative updates (MU) algorithm.*

Let $(h^{(t)})$ be a sequence generated by the MU algorithm.

- The objective function decreases at each iteration, and thus converges to a minimum.
- Let h^* be a limit point of the algorithm, such that $\forall k, h_k^* > 0$. Then h^* is a global minimum of l .

Theorem 5 *Speed of convergence.*

Let h^* be a global minimum of l . Suppose that one of the following conditions is met, for each $k \in \{1, \dots, K\}$:

-
-

Then, there exists $\delta > 0$ such that, for any $h^{(0)}$ such that $\|h^{(0)} - h^*\|_2 < \delta$, the sequence $(h^{(t)})$ generated by the MU algorithm converges to h^* at linear rate.

As we can see, the domain of the objective function can be decomposed into :

- basins of attraction, that is, open sets containing global minima, where linear convergence occurs.
- subsets of unclear nature
- absorbing boundary sets

Cases of sublinear convergence If there is one index k such that $h_k^* = 0$ then one of the following traps may occur :

•

Absorbing zero traps We will see on the following case that if $h_k^{(t)}$ gets too close to zero at some point in the sequence, then it may take some time for it to reach a linear convergence basin.

4.2 Mirror Descent Algorithm

(Hien and Gillis, 2021) derive another algorithm based on mirror descent. The authors make use of the fact that the objective function is smooth with respect to $\kappa(h) = -\log(h)$.

5 A modified MU algorithm

The mirror descent algorithm introduced in (Bauschke et al., 2017) is derived from an auxiliary function of the form $\sum_k \alpha_k \log(h_k) + \beta_k h_k$. A sublinear convergence rate is established for Poisson linear inverse problems, but it assumes that the optimum h^* of the problem is in the interior of the nonnegative cone.

We want to fill in the gap in the linear convergence rate established by Vincent et al. by solving the absorbing zero boundary problem : simply by introducing auxiliary functions of the form $\sum_k \alpha_k \log(\eta_k + h_k) + \beta_k h_k$.

5.1 Overview of the mu mod algorithm

$$h_k \leftarrow (h_k + \eta_k) \frac{\sum_f W_{fk} \frac{v_f}{\hat{v}_f}}{\sum_f W_{fk}}$$

$$h_k \leftarrow (h_k - \eta_k)_+$$

where $\hat{v} = Wh$ and $(x)_+ = \max(x, 0)$.

Parameter η_k plays two roles : firstly, choosing $\eta_k > 0$ suppresses the absorbing zero effect for coordinate k . Secondly, it acts as a threshold.

A difficult point : one must verify $\eta_k < h_k^*$ if we wish the algorithm to converge towards critical values of the objective function. On the other hand, if $h_k^* = 0$, $\eta_k > 0$ is interesting because of the thresholding effect.

A common value η for all coordinates may be chosen, which would simplify our presentation, but there might be some heuristics for choosing η_k , so we will pursue our analysis with a coordinate dependent value.

5.2 Convergence properties of the algorithm

We now explain the principle of the `mu mod` algorithm, and ask some questions regarding its tuning parameters.

We introduce the following parameterized family of functions :

$$U(h; z, q, c, \eta) = \sum_k (-z_k \log(\eta_k + h_k) + q_k h_k) + c$$

where $z, q, \eta \in \mathbb{R}^K$, and $c \in \mathbb{R}$.

For given values of z, γ, ϵ , the function U is defined on $\prod_k [-\eta_k, +\infty]$. It is convex as a function of h .

If all $\eta_k > 0$, then U is Lipschitz on $\prod_k [0, +\infty]$.

In order to pick the right members of this family, we will use the following lemma :

Theorem 6 *Let U be an auxiliary function for l at $\underline{h}$. Then $\nabla U(\underline{h}) = \nabla l(\underline{h})$*

This lemma is equivalent to saying that U is an auxiliary function for l at $\underline{h}$ if and only if $U - l$ reaches its minimum at $\underline{h}$.

The following conditions are necessary :

$q_k = \sum_f W_{fk}$ (asymptotic tangency at $h \rightarrow +\infty$)

$z_k = \eta_k + \underline{h}_k$ (tangency at $\underline{h}_k$)

$c = ..appropriateconstant...$

We consider the following claim : "for η sufficiently small, $h \mapsto U(h; z, q, c, \eta)$ is an auxiliary function for l , for all possible values of $\underline{h}$."

Clearly, if all η_k are positive, this is not true locally at $h \rightarrow 0$, since U is bounded whereas $l(h) \rightarrow +\infty$.

However, one could choose, at each iteration, a few "safety indices" such that $\eta_k = 0$, and others such that $\eta_k > 0$ in order to speed up convergence.

For instance, if one of the columns of W is the constant vector, or has all positive coefficients, it might be chosen as a safety index at all iterations.

6 Numerical experiments

Since we cannot guarantee global upper bounds, numerical experiments are needed to check whether the algorithm works frequently enough, and try simple backtracking rules.

6.1 A reference example in small dimension

$$V = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$$

$$W = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}$$

In this example, we can obtain an exact factorization $V = WH$ with :

$$H = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

In the following test, $H^{(0)}$ is initialized with the centered scaled method : $H^{(0)} = \begin{pmatrix} 2/3 & 2 & 8/3 \\ 1/3 & 1 & 4/3 \end{pmatrix}$.

We monitor below the objective function against the number of iterations. Multiplicative updates seem to converge at linear rate after a few tens of iterations, but yield only 10^{-5} after 1000 iterations, against 10^{-12} for the modified multiplicative updates.

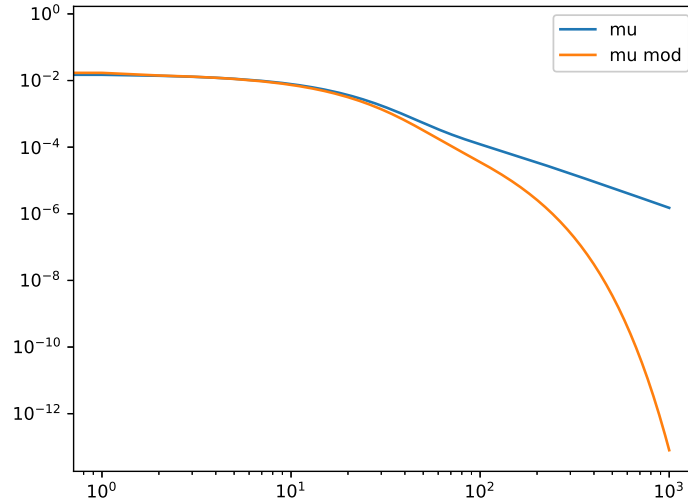


Figure 2: Comparison of `nmf mu` and `nm mu mod` on the `fan-exact` test case

6.2 [incomplete] Synthetic data : noiseless case

In the noiseless case, we have $V = WH$ for some unknown value H^* . If we assume that no column of W is a conic combination of the others, then H^* is the unique global minimum of our problem. Since we assume $K < F$, this is equivalent to assuming that W has full-rank. In that case, we can retrieve H^* by standard linear algebra.

Since we can only guarantee local upper-bounds for our auxiliary functions, it is interesting to search in which case we effectively obtain descent : cases where the bounds are invalid could help us devise better bounds, propose simple backtracking rules, etc.

W is drawn uniform in $[1, 2]$. The columns are then averaged to sum to 1.

We take $F = 50, N = 100, K = 40$.

$N = 100$ means that we average results over 100 samples of h .

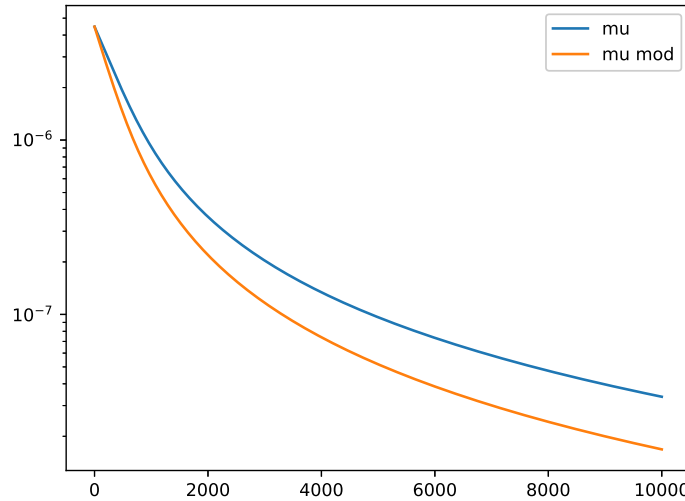


Figure 3: Comparison of `nmf mu` and `nm mu mod` on synthetic data : exact case

6.3 [incomplete] Synthetic data : gaussian noise

In this example, the coefficients of W are sampled from a gamma distribution with shape 100 and scale 1, and those of h from an exponential distribution of scale 1.

In the noisy case we can draw V from a Poisson distribution with scale WH .

We take $F = 50, N = 100, K = 40$. $N = 100$ means that we average results over 100 samples of h .

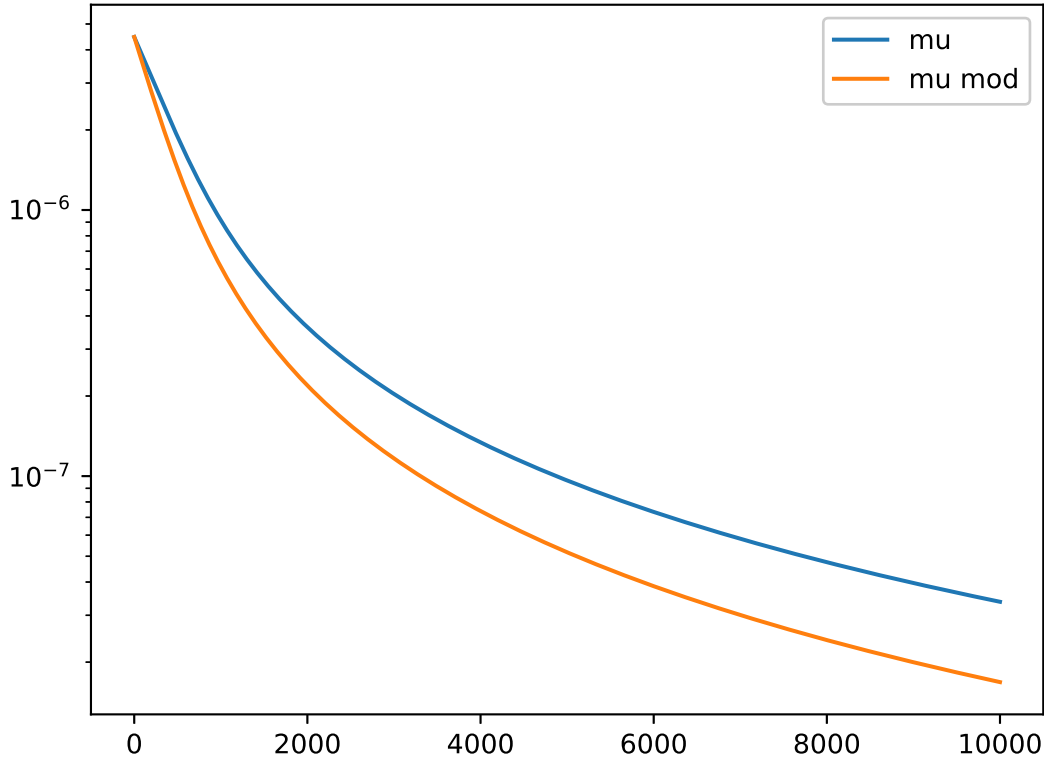


Figure 4: Comparison of nmf mu and nm mu mod on synthetic data : exact case

6.4 Synthetic data : Poisson noise

In order to get some statistical intuition, let us recall that F plays the role of the number of samples and K the dimension of the explanatory variables. W plays the role of the design matrix (usually written X in the statistical literature).

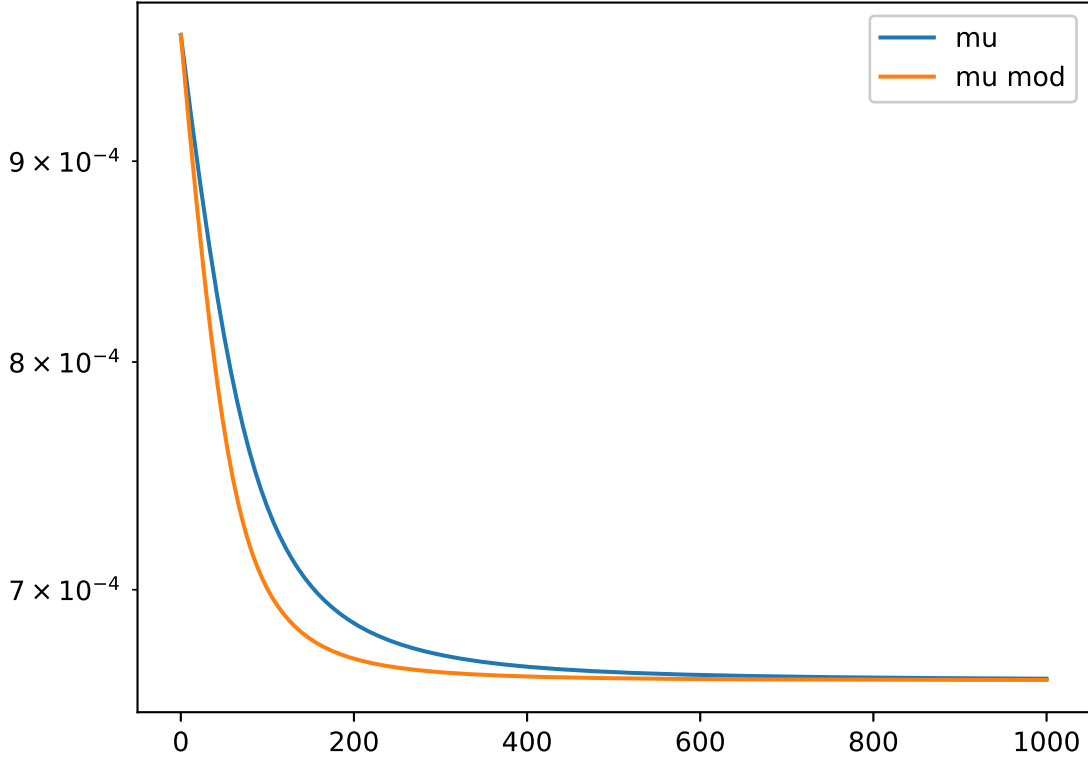


Figure 5: Comparison of `nmf mu` and `nm mu mod` on synthetic data : noisy case

In the noisy case, the true parameter H_0 is no longer a minimum of the objective function

In the large F limit, the true parameter used to simulate data should be a minimum of the objective function. Taking this into account, one would expect that in the large F limit, the objective function of our algorithms should converge to zero.

However, another effect comes into play : in the large F limit, the samples from the Poisson distribution get arbitrarily close to zero, so the condition number of our objective function become worse and worse.

7 Unsupervised KL-NMF : model and algorithm

7.1 Why overcomplete NMF ?

Intuitively, in the 3-dimensional case ($F = 3$), the cone section is a polygon in the perspective plane.

7.2 Model identification and hardness of the optimization problem

Another approach (Esser et al., 2012) is to constrain the dictionary entries to be combinations of endmembers. This leads to a convex optimization problem. However this is a strong hypothesis, and may lead to "overlearning" (*surapprentissage*).

7.3 Penalties for the cone

7.3.1 A recursive algorithm for $volume(W)$

We propose a recursive algorithm to compute the volume of $\mathcal{C}(W)$.

- Choose vertex K and retrieve the $F - 1$ nearest neighbours. Together they form $W' \in \mathbb{R}_+^{F \times F}$. Compute $V' = volume(W') = \det(W')$.
- Let W'' be the matrix formed by removing vertex $K - 1$.

7.3.2 penalties for the spread $\mathcal{C}(W)$

Let $w = \sum_{k \in range(K)} W[:, k]$. Since all vertices lie in the perspective hyperplane, minimum volume occurs when the vertices get closer and closer to the center. Instead of the volume, one may compute the spread of $\mathcal{C}(W)$, in one of several manners :

1. $\sum_{k \in range(K)} \|W[:, k] - w\|_2^2$: a smooth penalty function
2. $\sum_{k \in range(K)} \|W[:, k] - w\|_2$: a nonsmooth penalty function with a sparsity like feeling.
3. $\max_{k \in range(K)} \|W[:, k] - w\|_2$: something like a half-diameter. Although it is nonsmooth, it might incur less "shape stressing" since it only stresses the remotest vertex.

8 Conclusion

We have proposed an algorithm in the underdetermined case. Clearly, the ability of our algorithm to handle zero coefficients and suppress the absorbing zero effect makes it a good choice for the overdetermined case. Moreover, sparsity-promoting penalty terms can naturally be incorporated in our model (Bauschke et al., 2017).

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